



MAX-PLANCK-GESELLSCHAFT

Stochastic Fitzhugh-Nagumo neuron model in excitable regime embeds a leaky-integrate-and-fire model

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FHN deterministic model

In the fast time scale t , the deterministic FHN neuron model is

$$\begin{cases} dv_t &= (v_t - \frac{v_t^3}{3} - w_t + I)dt = f(v_t, w_t)dt, \\ dw_t &= \varepsilon(v_t + \alpha - \beta w_t)dt = g(v_t, w_t)dt, \end{cases} \quad (1.1)$$

where $0 < \varepsilon := t/\tau \ll 1$.

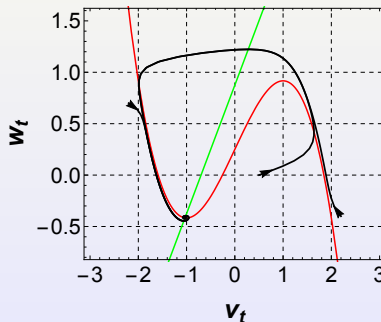


Figure: stable fixed point $(v_e, w_e) = (-1.03248, -0.4156)$. $I = 0.25, \alpha = 0.7, \beta = 0.8, \varepsilon = 0.1$.

Stochastic FHN model

We consider this stochastic FHN model

$$\begin{cases} dv_t &= f(v_t, w_t)dt, \\ dw_t &= g(v_t, w_t)dt + h(w_t) \circ dB_t, \end{cases} \quad (1.2)$$

where $h(w) = \sigma_0$ (additive channel noise) or $h(w) = \sigma_0 w$ (multiplicative channel noise).

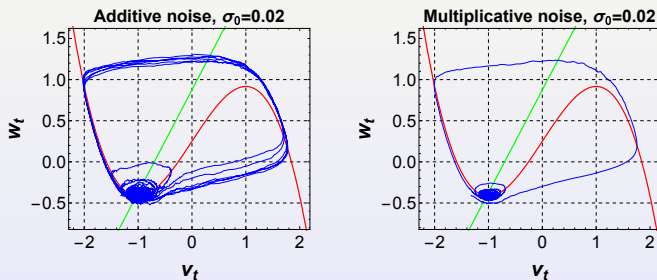


Figure: Random trajectory of Eq. (1.2) in the excitable regime ($I = 0.25, \alpha = 0.7, \beta = 0.8, \varepsilon = 0.1$) with the initial condition $(v_0, w_0) = (-1.32, -0.41)$ for both additive and multiplicative noise ($T = 2000$).

Spiking frequency

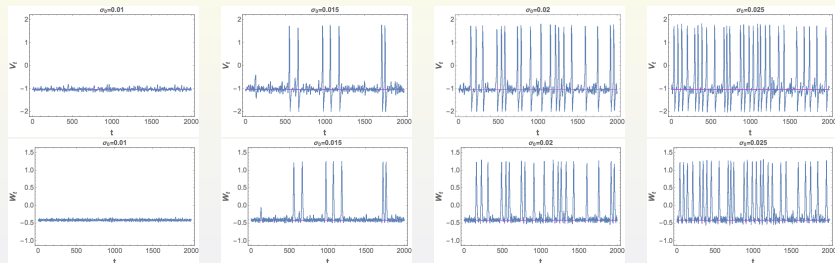


Figure: For certain noise amplitudes, the trajectory can escape from the basin of attraction of the globally stable equilibrium point at $(v_e, w_e) = (-1.03248, -0.4156)$, thereby inducing spiking. For a small noise amplitude $\sigma_0 = 0.01$ there is no spiking until time $t = 2000$, and the larger the noise, the more spikes we have. $\sigma_0 \in \{0.01, 0.015, 0.02, 0.025\}$. Left column: additive noise; Right column: multiplicative noise.

Stochastic FHN

Let $\mathbf{X} = (v, w)^T$ and $F(\mathbf{X}), H(\mathbf{X}) \in \mathbb{R}^2$ be the drift and diffusion coefficients of (1.2). The stochastic system is then of the form

$$d\mathbf{X}_t = F(\mathbf{X}_t)dt + H(\mathbf{X}_t) \circ dB_t, \quad (1.3)$$

where

$$H(\mathbf{X}) = (0, \sigma_0)^T, \quad \text{or} \quad H(\mathbf{X}) = B\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \end{pmatrix} \mathbf{X}, \quad \text{or} \quad H(\mathbf{X}) = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \frac{w}{w^2+1} \end{pmatrix} \quad (1.4)$$

and $\circ dB_t$ stands for the Stratonovich stochastic integral with respect to the Brownian motion B_t ; or dB_t^H - the stochastic differential w.r.t. a fractional Brownian motion. It is easy to check that F is dissipative in the weak sense, i.e.

$$\begin{aligned} \langle \mathbf{X}_1 - \mathbf{X}_2, F(\mathbf{X}_1) - F(\mathbf{X}_2) \rangle &= (v_1 - v_2)^2 \left[1 - \frac{1}{3}(v_1^2 + v_1 v_2 + v_2^2) \right] \\ &\quad - (1 - \epsilon)(v_1 - v_2)(w_1 - w_2) - \epsilon\beta(w_1 - w_2)^2 \\ &\leq 3 \left(1 + \frac{\epsilon\beta}{2} + \frac{(1 - \epsilon)^2}{2\epsilon\beta} \right)^2 - \frac{\epsilon\beta}{2} \|\mathbf{X}_1 - \mathbf{X}_2\|^2 \end{aligned} \quad (1.5)$$

Stationary distribution

Markov generator:

$$\mathcal{A}p = f \frac{\partial p}{\partial v} + g \frac{\partial p}{\partial w} + \frac{1}{2} \sum_{i,j=1}^2 [HH^T]_{ij} \frac{\partial^2 p}{\partial v_i \partial w_j}$$

Mattingly, Stuart, Higham (SPA 2002)- Lyapunov function: there exists a function $V(\mathbf{X}) = \|\mathbf{X}\|^{2l}$ and a $c_l, d_l > 0$ such that

$$\mathcal{A}V(\mathbf{X}) \leq d_l - c_l V(\mathbf{X})$$

Hoermander condition: the solution p is smooth.

There exists a unique stationary distribution π s.t. for any $|q(\mathbf{X})| \leq 1 + \|\mathbf{X}\|^{2l}$

$$|E^{\mathbf{X}_0} q(\mathbf{X}_t) - \pi(q)| \leq \kappa(1 + \|\mathbf{X}_0\|^{2l})e^{-\lambda t}.$$

implies that π is point measure attractor! But in practice, we sometimes need more: pointwise convergence.

Random dynamical systems

$(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which our Brownian motion B_t is defined. Ω can be chosen as $C^0(\mathbb{R}, \mathbb{R})$, the space of continuous real functions on $\mathbb{R}$ which are zero at zero, equipped with the compact open topology in $\mathbb{R}$, $\mathcal{F}$ as $\mathcal{B}(C^0)$, the associated Borel- σ -algebra and $\mathbb{P}$ as the Wiener measure. The Brownian motion B_t can then be constructed as the canonical version $B_t(\omega) := \omega(t)$.

On this probability space we construct a dynamical system θ as the Wiener shift

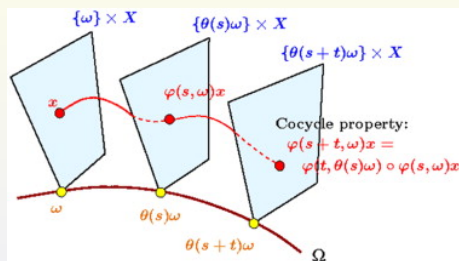
$$\theta_t m(\cdot) = m(t + \cdot) - m(t), \quad \forall t \in \mathbb{R}, \forall m \in \tilde{\Omega}. \quad (2.1)$$

Then $\theta_t(\cdot) : \Omega \rightarrow \Omega$ satisfies the group property, i.e. $\theta_{t+s} = \theta_t \circ \theta_s$ for all $t, s \in \mathbb{R}$, and is $\mathbb{P}$ -preserving, i.e. $\mathbb{P}(\theta_t^{-1}(A)) = \mathbb{P}(A)$ for every $A \in \mathcal{F}$, $t \in \mathbb{R}$. The quadruple $((\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in \mathbb{R}}))$ is called a *metric dynamical system*.

A (continuous) random dynamical system is an NDS $(t, \omega, x) \mapsto \varphi(t, \omega)x$ is measurable w.r.t $\mathcal{B} \otimes \mathcal{F} \otimes \mathcal{B}(X)$ and $\mathcal{B}(X)$, respectively, where $\mathcal{B}(X)$ is the Borel σ -algebra of X , such that the family $\varphi(t, \omega, \cdot) = \varphi(t, \omega) : X \rightarrow X$ of self-mappings of X satisfies the *cocycle property*

$$\varphi(0, \omega) = id_X, \quad \varphi(t + s, \omega) = \varphi(t, \theta_s \omega) \circ \varphi(s, \omega), \quad \forall t, s \in T, \quad \forall \omega \in \Omega. \quad (2.2)$$

Generation of RDS



- SDEs with standard Brownian motions: L. Arnold, P. Kloeden, T. Caraballo, B. Schmalfuss, H. Crauel, M. Scheutzow...

$$dX_t = f(X_t)dt + g(X_t)dW_t. \quad (2.3)$$

- SDEs with fractional Brownian motions: M. Hairer, B. Schmalfuss, M. Scheutzow,...(also with SPDE)

$$dX_t = f(X_t)dt + g(X_t)dB_t^H, H \in (0, 1). \quad (2.4)$$

Given $H \in (0, 1)$, a continuous centered Gaussian process $B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s, t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}), \quad s, t \in \mathbb{R},$$

on an appropriate probability space, is called a two-sided one-dimensional fractional Brownian motion (fBm) with Hurst parameter H . When $H = 1/2$, B is the standard Brownian motion.

Fractional Brownian motion B^H using the representation in Mandelbrot and van Ness

$$B^H(t) := \frac{1}{c_H} \left(\int_{-\infty}^0 [(t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}}] dW(u) + \int_0^t (t-u)^{H-\frac{1}{2}} dW(u) \right),$$

with

$$c_H := \left(\frac{1}{2H} + \int_0^\infty [(1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}}]^2 du \right)^{\frac{1}{2}}.$$

Young integrals

$0 < \alpha < 1$, $C^{\alpha-\text{Hol}}([T_1, T_2], \mathbb{R}^d)$ the space of Hölder continuous functions on $[T_1, T_2]$

$$\|u\|_{\alpha, [T_1, T_2]} := |u_{T_1}| + \sup_{T_1 \leq s < t \leq T_2} \frac{|u(t) - u(s)|}{(t - s)^\alpha}.$$

We also denote by $\|\cdot\|_{\infty, [T_1, T_2]}$ the supnorm of continuous functions on $[T_1, T_2]$.

Now, consider $x \in C^\alpha([T_1, T_2], \mathbb{R}^{d \times m})$ and $\omega \in C^\beta([T_1, T_2], \mathbb{R}^m)$, $\alpha + \beta > 1$.

Riemann-Stieltjes sums for finite partition $\Pi = \{T_1 = t_0 < t_1 < \dots < t_n = T_2\}$ of $[T_1, T_2]$ and any $\xi_i \in [t_i, t_{i+1}]$

$$S_\Pi := \sum_{i=0}^{n-1} x(\xi_i)(\omega(t_{i+1}) - \omega(t_i)),$$

converges as the mesh $|\Pi| := \max_{0 \leq i \leq n-1} |t_{i+1} - t_i|$ tends to zero, we call the limit is the

Young integral of x w.r.t ω on $[T_1, T_2]$ denoted by $\int_{T_1}^{T_2} x(t) d\omega(t)$. Also Young-Loeve estimate

$$\left| \int_s^t x(u) d\omega(u) - x(s)[\omega(t) - \omega(s)] \right| \leq K |t - s|^{\alpha+\beta} \|x\|_{\alpha, [s, t]} \|\omega\|_{\beta, [s, t]}, \quad (2.5)$$

where $K := (1 - 2^{1-\theta})^{-1}$, $\theta := \alpha + \beta > 1$.

Random attractors

Random sets: $\hat{M} = \{M(\omega)\}_{\omega \in \Omega}$ such that $\omega \mapsto d(x|M(\omega))$ is $\mathcal{F}$ -measurable for each $x \in X$. Universe: $\mathcal{D}$ a family of random sets which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$).

- Invariance: $\varphi(t, \omega)A(\omega) = A(\theta_t \omega)$
- Pullback random attractor: $\lim_{t \rightarrow \infty} d(\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega)|A(\omega)) = 0$;
- Forward random attractor: $\lim_{t \rightarrow \infty} d(\varphi(t, \omega)D(\omega)|A(\theta_t \omega)) = 0$.

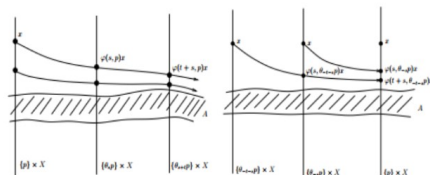


Figure 2.1: Forward attraction versus pullback attraction.

Random attractors for dissipative systems

$$d\mathbf{X}_t = F(\mathbf{X}_t)dt + H(\mathbf{X}_t) \circ dB_t, \quad (2.6)$$

where

$$H(\mathbf{X}) = (0, \sigma_0)^T, \quad \text{or} \quad H(\mathbf{X}) = B\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \end{pmatrix} \mathbf{X}, \quad \text{or} \quad H(\mathbf{X}) = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \frac{w}{w^2+1} \end{pmatrix} \quad (2.7)$$

and $\circ dB_t$ stands for the Stratonovich stochastic integral with respect to the Brownian motion B_t ; or dB_t^H - the stochastic differential w.r.t. a fractional Brownian motion.

Theorem

There exists a random attractor for (2.6).

$H = \frac{1}{2}$: Arnold & Schmalfuss (1996), Crauel & Flandoli (1994)

$H \neq \frac{1}{2}$: Garrido-Atienza & Neuenkirch & Kloeden (2009), Gess et al. (2011) for SPDE with additive noise...

Idea: transform the SDE into a conjugated RDE and then prove that there exists an absorbing set.

Existence of random attractor

Pullback absorbing set: $\hat{B} \in \mathcal{D}$ in a universe $\mathcal{D}$ such that $\hat{B}$ absorbs all sets in $\mathcal{D}$, i.e. for any $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega) \subset B(\omega), \text{ for all } t \geq t_0.$$

Theorem (Schenk-Hoppe 1998: Existence and Uniqueness of Random Attractor)

Given a universe $\mathcal{D}$, assume there exists a random compact pullback absorbing set $\hat{B} \in \mathcal{D}$ which is forward invariant. Then there exists a unique random pullback attractor (which is then a weak attractor) in $\mathcal{D}$, given by

$$A(\omega) = \bigcap_{t \geq 0} \varphi(t, \theta_{-t}\omega)B(\theta_{-t}\omega).$$

Application: prove that a system has an absorbing set.

Sketch of the proof

Multiplicative noise: introduce the transformation $\mathbf{X}_t = e^{\sigma Z_t} \mathbf{Y}_t$ where z_t is the unique stationary solution of the Langevin equation

$$dz_t = -z_t dt + dB_t.$$

By the integration by part rule, it follows that

$$d\mathbf{X}_t = \sigma \mathbf{X}_t dz_t + e^{\sigma Z_t} \mathbf{Y}_t,$$

therefore system (2.6) will be transformed into a random differential equation

$$\dot{\mathbf{Y}}_t = g(z_t, \mathbf{Y}_t) := e^{-\sigma Z_t} f(e^{\sigma Z_t} \mathbf{Y}_t) + \sigma z_t \mathbf{Y}_t. \quad (2.8)$$

Observe that

$$\begin{aligned} \frac{d}{dt} \|\mathbf{Y}_t\|^2 &= 2\langle \mathbf{Y}_t, g(z_t, \mathbf{Y}_t) \rangle \\ &= 2e^{-2\sigma Z_t} \langle e^{\sigma Z_t} \mathbf{Y}_t, f(e^{\sigma Z_t} \mathbf{Y}_t) + 2\sigma z_t \|\mathbf{Y}_t\|^2 \rangle \\ &\leq 2e^{-2\sigma Z_t} [a - b \|e^{\sigma Z_t} \mathbf{Y}_t\|^2 + \langle f(0), e^{\sigma Z_t} \mathbf{Y}_t \rangle] + 2\sigma z_t \|\mathbf{Y}_t\|^2 \\ &\leq \left[2a + \frac{\|f(0)\|^2}{b} \right] e^{-2\sigma Z_t} + (2\sigma z_t - b) \|\mathbf{Y}_t\|^2 =: p(z_t) + q(z_t) \|\mathbf{Y}_t\|^2. \end{aligned}$$

Hence by the comparison principle, $\|\mathbf{Y}_t\|^2 \leq R_t$ whenever $\|\mathbf{Y}_0\|^2 \leq R_0$ where R_t is the solution of

$$\dot{R}_t = p(z_t) + q(z_t)R_t, \quad (2.9)$$

which can be computed explicitly as

$$R_t(\omega, R_0) = e^{\int_0^t q(z_u(\omega)) du} R_0 + \int_0^t p(z_s(\omega)) e^{\int_s^t q(z_u(\omega)) du} ds.$$

Observe that by the Birkhorff ergodic theorem, there exist almost surely

$$\lim_{t \rightarrow -\infty} \frac{1}{t} \int_t^0 q(z_u) du = \lim_{t \rightarrow -\infty} \frac{1}{t} \int_t^0 q(z(\theta_u \omega)) = E q(z(\cdot)) = -b < 0,$$

hence there exists a unique stationary solution of (2.9) which can be written in the form

$$\bar{R}(\omega) = \int_{-\infty}^0 p(z_s(\omega)) e^{\int_s^0 q(z_u(\omega)) du} ds.$$

Moreover $\|\mathbf{Y}_t(\omega, \mathbf{Y}_0)\|^2 \leq \bar{R}(\theta_t \omega)$ whenever $\|\mathbf{Y}_0\|^2 \leq \bar{R}(\omega)$ and

$$\limsup_{t \rightarrow \infty} \|\mathbf{Y}_t(\theta_{-t} \omega, \mathbf{Y}_0)\|^2 \leq \limsup_{t \rightarrow \infty} R_t(\theta_{-t} \omega, R_0) = \bar{R}(\omega).$$

Hence the ball $B(0, R(\omega))$ is actually forward invariant under the random dynamical system generated by (2.8) and is also a pullback absorbing set. Thus there exists a random attractor for (2.8). Finally, since systems (2.6) and (2.8) are conjugate under the random linear transformation $\mathbf{X}_t = e^{\sigma z_t} \mathbf{Y}_t$ which is tempered, there exists also a random attractor for system (2.6).

Additive noise: In this case define $\mathbf{Y}_t = \mathbf{X}_t - \eta_t$ where η_t is the unique stationary solution of

$$d\eta_t = -\eta_t dt + \sigma dB_t$$

System (2.6) is then transformed to

$$\dot{\mathbf{Y}}_t = f(\mathbf{Y}_t + \eta_t) + \eta_t. \quad (2.10)$$

Observe that

$$\begin{aligned} \frac{d}{dt} \|\mathbf{Y}_t\|^2 &= 2\langle \mathbf{Y}_t, f(\mathbf{Y}_t + \eta_t) - f(\eta_t) \rangle + 2\langle \mathbf{Y}_t, f(\eta_t) + \eta_t \rangle \\ &\leq 2(a - b\|\mathbf{Y}_t\|^2) + b\|\mathbf{Y}_t\|^2 + \frac{1}{b}\|f(\eta_t) + \eta_t\|^2 \\ &\leq 2a + \frac{1}{b}\|f(\eta_t) + \eta_t\|^2 - b\|\mathbf{Y}_t\|^2. \end{aligned}$$

By the same argument as the first case, there exists a random radius

$$R^*(\omega) = \int_{-\infty}^0 [2a + \frac{1}{b} \|f(\eta_s) + \eta_s\|^2] e^{bs} ds,$$

such that $\mathbf{X}_t(\omega, \mathbf{X}_0) \in B(\eta_t, R^*(\theta_t\omega))$ whenever $\mathbf{X}_0 \in B(\eta_0, R^*(\omega))$ by the comparison principle, and furthermore

$$\limsup_{t \rightarrow \infty} \|\mathbf{Y}_t(\theta_{-t}\omega, \mathbf{Y}_0)\|^2 \leq \limsup_{t \rightarrow \infty} R_t(\theta_{-t}\omega, R_0) = R^*(\omega).$$

Hence the random ball $B(\eta, R^*)$ is a forward invariant pullback absorbing set of the random dynamical system generated by (2.6). Similar arguments then show that there exists a random attractor for (2.6).

Normal form

we shall study the dynamics of (1.2) in a small vicinity of the fixed point $\mathbf{X}_e = (v_e, w_e)$. To do that, consider the shift system w.r.t. the fixed point $\mathbf{X}_e$ which has the form

$$\begin{aligned} d(\mathbf{X}_t - \mathbf{X}_e) &= [F(\mathbf{X}_t) - F(\mathbf{X}_e)]dt + H(\mathbf{X}_t) \circ dB_t \\ &= \left[DF(\mathbf{X}_e)(\mathbf{X}_t - \mathbf{X}_e) + \bar{F}(\mathbf{X}_t - \mathbf{X}_e) \right] dt + H(\mathbf{X}_t) \circ dB_t, \end{aligned} \quad (3.1)$$

with initial point $\mathbf{X}_0 - \mathbf{X}_e$, where $DF(\mathbf{X}_e)$ is the linearized matrix of F at $\mathbf{X}_e$, $\bar{F}$ is the nonlinear term such that

$$\|\bar{F}(\mathbf{X} - \mathbf{X}_e)\| = \left\| \begin{pmatrix} \frac{1}{3}|v + 2v_e|(v - v_e)^2 \\ 0 \end{pmatrix} \right\| \leq \gamma(r)\|\mathbf{X} - \mathbf{X}_e\|, \quad \forall \|\mathbf{X} - \mathbf{X}_e\| \leq r$$

for an increasing function $\gamma(\cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $r \mapsto \frac{r^2}{3} + |v_e|r$, which implies that $\lim_{r \rightarrow 0} \gamma(r) = 0$. Since $H(\mathbf{X})$ is either a constant or a linear function, we prove below that system (3.1) can be well approximated by its linearized system

$$d\bar{\mathbf{X}}_t = DF(\mathbf{X}_e)\bar{\mathbf{X}}_t dt + H(\bar{\mathbf{X}}_t + \mathbf{X}_e) \circ dB_t, \quad \bar{\mathbf{X}}_0 = \mathbf{X}_0 - \mathbf{X}_e. \quad (3.2)$$

Theorem

Given $\|\mathbf{X}_0 - \mathbf{X}_e\| < r$ and equations (3.1), (3.2), define the stopping time $\tau = \inf\{t > 0 : \|\mathbf{X}_t - \mathbf{X}_e\| \geq r\}$. Then there exists a constant C independent of r such that for any $t \geq 0$, the following estimates hold

- For additive noise $\sup_{t \leq \tau} \|\mathbf{X}_t - \mathbf{X}_e - \bar{\mathbf{X}}_t\| \leq C\gamma(r)r$.
- For multiplicative noise $E\|\mathbf{X}_{t \wedge \tau} - \mathbf{X}_e - \bar{\mathbf{X}}_{t \wedge \tau}\|^2 \leq C\gamma^2(r)r^2$.

Comments:

$$DF(\mathbf{X}_e) = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}$$

has two conjugate complex eigenvalues with negative real part

$$\lambda_{1,2} = \frac{1}{2}(1 - \nu_e^2 - \epsilon\beta) \pm \frac{i}{2}\sqrt{4\epsilon - (1 - \nu_e^2 + \epsilon\beta)^2} = -0.0730077 \pm 0.31615i = -\mu \pm \nu i.$$

Hence by using the transformation $\mathbf{X} - \mathbf{X}_e = Q\mathbf{Y}$ and $\bar{\mathbf{X}} = Q\bar{\mathbf{Y}}$ with

$$Q = \begin{pmatrix} -\nu & m_{11} + \mu \\ 0 & m_{21} \end{pmatrix},$$

Additive noise: $C := \frac{1}{\mu}\|Q\|\|Q^{-1}\|$. Multiplicative noise: $C := \frac{1}{\mu} \frac{1}{\lambda}\|Q^{-1}\|^2\|Q\|^2$

LIF model

The first method constructs the so-called *radial Ornstein-Uhlenbeck equation*. More precisely, we rewrite the linearized system (3.2) in the form

$$d\bar{\mathbf{X}}_t = DF(\mathbf{X}_e)\bar{\mathbf{X}}_t dt + \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \end{pmatrix} d\mathcal{B}_t, \quad \mathcal{B}_t = \begin{pmatrix} B'_t \\ B_t \end{pmatrix} \quad (3.3)$$

By transformation $\bar{\mathbf{Y}}_t = Q^{-1}\bar{\mathbf{X}}_t$ we obtain

$$d\bar{\mathbf{Y}}_t = A\bar{\mathbf{Y}}_t dt + C d\mathcal{B}_t, \quad A = \begin{pmatrix} -\mu & \nu \\ -\nu & -\mu \end{pmatrix}; C = Q^{-1} \begin{pmatrix} 0 & 0 \\ 0 & \sigma_0 \end{pmatrix}. \quad (3.4)$$

We note that $\frac{\mu}{\nu}$ is small, therefore, by applying the technique of time average,

$$\bar{\mathbf{Y}}_t \sim \bar{\mathbf{Y}}_t^{app} := \frac{\sigma}{\sqrt{\mu}} Rot_{-\nu t} \bar{\mathbf{S}}_{\mu t}, \quad \sigma = \sqrt{\frac{1}{2} \text{tr}(CC^*)} = \sqrt{\frac{-m_{12}}{2\nu^2 m_{21}}} \sigma_0$$

where $Rot_s := \begin{pmatrix} \cos s & -\sin s \\ \sin s & \cos s \end{pmatrix}$, and $d\bar{\mathbf{S}}_t = -\bar{\mathbf{S}}_t dt + d\mathcal{B}_t$, with the initial value

$\bar{\mathbf{S}}_0 = \frac{\sqrt{\mu}}{\sigma} \bar{\mathbf{Y}}_0$. Therefore, $\|\bar{\mathbf{Y}}_t\|$ can be approximated by $R_t := \|\bar{\mathbf{Y}}_t^{app}\| = \frac{\sigma}{\sqrt{\mu}} \|\bar{\mathbf{S}}_{\mu t}\|$

$$dR_t = \left[\frac{\sigma^2}{2R_t} - \mu R_t \right] dt + \sigma d\tilde{B}_t. \quad (3.5)$$

The second method is to consider $\bar{\mathbf{Y}}_t$ in polar coordinates with

$$d\bar{\mathbf{Y}}_t = A\bar{\mathbf{Y}}_t dt + {}_e dB_t,$$

where ${}_e = Q^{-1} \begin{pmatrix} 0 \\ \sigma_0 \end{pmatrix} = \begin{pmatrix} 0.22117\sigma_0 \\ 10\sigma_0 \end{pmatrix}$. Its norm $\bar{R}_t := \|\bar{\mathbf{Y}}_t\|$ and its angle $\theta_t = \frac{\bar{\mathbf{Y}}_t}{\bar{R}_t}$ satisfy

$$\begin{aligned} d\bar{R}_t &= \left[\frac{\|{}_e\|^2 - \langle {}_e, \theta_t \rangle^2}{2\bar{R}_t} - \mu\bar{R}_t \right] dt + \langle \theta_t, {}_e \rangle dB_t, \\ d\theta_t &= \left[(A + \mu I)\theta_t - \frac{\|{}_e\|^2 - \langle {}_e, \theta_t \rangle^2}{2\bar{R}_t^2} \theta_t \right] dt + \frac{1}{\bar{R}_t} \left[{}_e - \langle {}_e, \theta_t \rangle \theta_t \right] dB_t. \end{aligned}$$

By the averaging technique, one can approximate $\theta_t = \begin{pmatrix} \sin \nu t \\ \cos \nu t \end{pmatrix}$, hence

$$\begin{aligned} d\bar{R}_t &= \left[\frac{100.049\sigma_0^2 - (0.22117 \sin \nu t + 10 \cos \nu t)^2 \sigma_0^2}{2\bar{R}_t} - \mu\bar{R}_t \right] dt \\ &\quad + (0.22117 \sin \nu t + 10 \cos \nu t) \sigma_0 dB_t. \end{aligned} \quad (3.6)$$

A spike in Eq. (1.2) occurs when there is a transition of a random trajectory from the vicinity of the stable fixed point $\mathbf{X}_e = (v_e, w_e)$ located on the left stable part of $\mathcal{C}_0$ to its right stable part and back to the vicinity of $\mathbf{X}_e$. This spike happens almost surely when a random trajectory with the starting point $\mathbf{X}_0$ in the vicinity of $\mathbf{X}_e$ crosses the threshold line $w = 0.5$.

The conditional probability of firing should be of the form

$$p(l) = \frac{1}{1 + e^{\frac{\alpha - l}{\beta}}}. \quad (3.7)$$

The parameters α and β then are estimated by using a non-linear regression from the above simulation data and are plotted in Fig. 4 for different values of the noise amplitude σ_0 .

To simplify calculations we will work on the transformed coordinates $\bar{\mathbf{Y}}_t$. Then the distance l between $(l, 0)$ and $(0, 0)$ in $\bar{\mathbf{X}}_t$ transforms to the distance

$$r = \left| Q^{-1} \begin{pmatrix} l \\ 0 \end{pmatrix} \right| = \frac{l}{\nu}$$

and the conditional probability of firing Eq. (3.7) transforms to

$$p(r) = \frac{1}{1 + e^{\frac{\alpha^* - r}{\beta^*}}}, \quad \alpha^* = \frac{\alpha}{\nu}, \quad \beta^* = \frac{\beta}{\nu}. \quad (3.8)$$

Firing mechanism

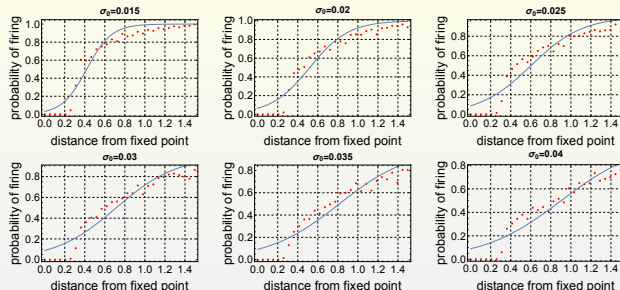


Figure: Conditional probability of spiking when crossing the line $L = \{(v, w_e) : v_e \leq v\}$ for different values of the noise amplitude σ_0

σ_0	0.015	0.02	0.025	0.03	0.035	0.04	0.045
α	0.419196	0.520319	0.602628	0.714283	0.814938	0.903889	0.97034
β	0.123239	0.213248	0.257344	0.306029	0.357886	0.390891	0.43386
α^*	1.32594	1.6458	1.90614	2.25931	2.57769	2.85905	3.0692
β^*	0.389811	0.674514	0.813992	0.967985	1.13201	1.23641	1.3723

ISI distribution

Let τ_1 be the first firing time. We computed the density of the distribution of τ_1 in terms of the conditional hazard rate

$$\alpha(r, t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(t \leq \tau_1 < t + \Delta t | \tau_1 \geq t, R_t = r).$$

From standard results from survival analysis, the density of the firing time

$$g(t) = \frac{d}{dt} P(\tau_1 \leq t) = E \left(\alpha(R_t) e^{-\int_0^t \alpha(R_s) ds} \right). \quad (3.9)$$

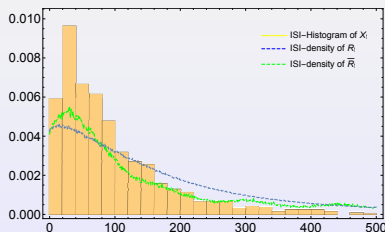


Figure: Compare ISI-histogram of FHN $\mathbf{X}_t$ and ISI-density of R_t and ISI-density of $\bar{R}_t$ with constructed firing mechanism, $\sigma_0 = 0.02$.